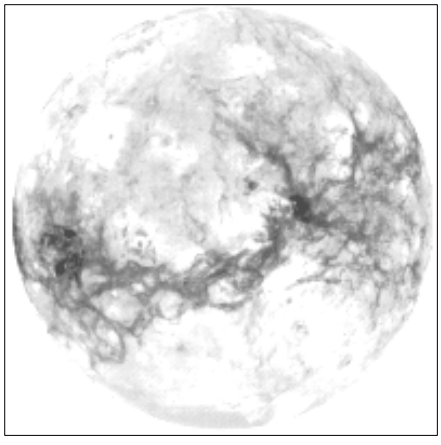
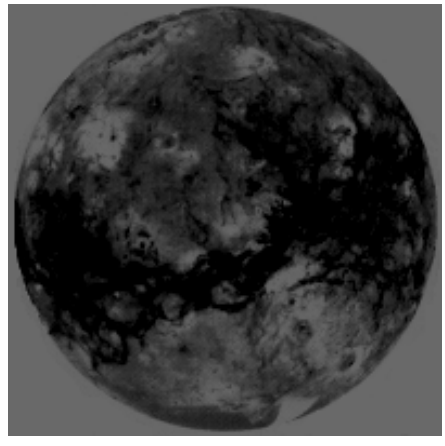
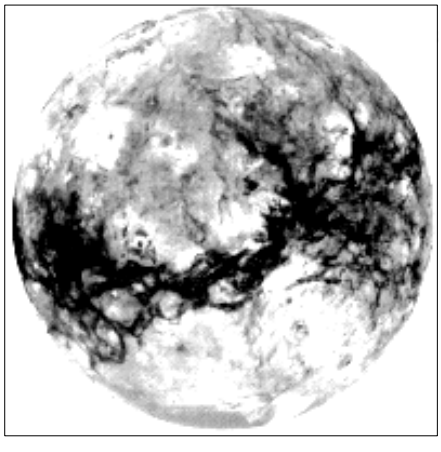


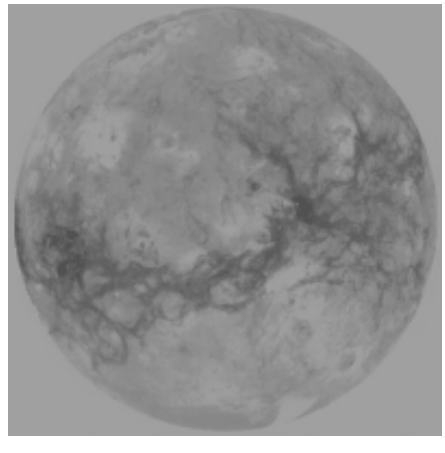
a. Brightness too high



b. Brightness too low



c. Contrast too high



d. Contrast too low

FIGURE 23-10
Brightness and contrast adjustments. Increasing the *brightness* makes every pixel in the image becomes lighter. In comparison, increasing the *contrast* makes the light areas become lighter, and the dark areas become darker. These images show the effect of misadjusting the brightness and contrast.

the contrast set to high, resulting in the blacks being too black, and the whites being too white. Lastly, (d) has the contrast set too low; all of the pixels are a mid-shade of gray making the objects fade into each other.

Figures 23-11 and 23-12 illustrate *brightness* and *contrast* in more detail. A test image is displayed in Fig. 23-12, using six different brightness and contrast levels. Figure 23-11 shows the construction of the test image, an array of 80×32 pixels, with each pixel having a value between 0 and 255. The backgound of the test image is filled with random noise, uniformly distributed between 0 and 255. The three square boxes have pixel values of 75, 150 and 225, from left-to-right. Each square contains two triangles with pixel values only slightly different from their surroundings. In other